Quality choice with reputation effects: Evidence from hospices in California

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Motivation

Hospices provide palliative care to dying patients.

1. Large industry, deficiencies in quality.
2. Insightful for regulated-price healthcare markets.
3. Quality choice under reputation effects.
    - Reputation of a firm reflects its past quality choices.
    - Current quality $\uparrow \implies$ reputation $\uparrow \implies$ future sales $\uparrow$

Develop structural model of hospice industry to show:

1. Hospices choose quality to build reputation, gain market share.
2. Counterfactual policies $\implies$ improve hospice quality and lower Medicare spending.

Hospice industry

1. Serve dying patients at residences.
    - Regular visits for pain-control, living arrangements.
2. Patient's choice aided by social worker/physician/community.
3. More visits $\implies$ higher quality.
    - Hospice is checking up on patients more and adjusting for day-to-day difficulties.
    - Hospice care is relatively low-skill.
    - Total visits proxies general **effort** by the hospice, since very costly.
    - Measure of hospice quality = Average visits-per-patient-day.

Hospice industry

4. No price competition: Paid fixed rate per-patient per-day by Medicare.
    - Most patients covered by Medicare + Medi-Caid.
5. Hospice reputation $\rightarrow$ patient's choice.
    - Quality not contractible: hospice unilaterally decides visits.
    - Goodwill and name recognition.
    - Good track record $\implies$ better known and suggested $\implies$ higher market share.

Data

Estimate structural model of i) demand for hospices and ii) reputation competition between hospices.

Data on hospices in California :

- Panel data on hospices at the **firm level**, 2002-2018.
- For each hospice-year: total patients served, total visits made by staff, hospice characteristics, mean patient-length-of-stay, etc.
- Combine with: data on population sizes (by age), mortality rates, and Medicare hospice reimbursement rates.
- Market defined at county level (28 counties).

Measure of hospice quality = visits-per-patient-per-day

Structural model

Demand: Every consumer chooses from set of hospices in her market.

- Choice depends on: hospice characteristics and hospice reputations.
- Essentially free for patient.
- **Reputation of a hospice = stock of its current and past quality choices, partially depreciates every period.**

Structural model

Supply: Hospices choose quality every year.

- Dynamic oligopoly model.
- Dynamic: Current quality choice $\rightarrow$ reputation $\rightarrow$ future market share.
- Oligopoly: Competing on reputation with other hospices.
- Trade-off: quality $\uparrow \implies$ market share $\uparrow$ and cost $\uparrow$.

Results: Estimation

Demand estimation:

1. Reputation plays important role in consumer choice.
   - Hospices which chose high quality **in the past** have higher **current** market share.
2. Reputation depreciates at annual rate of 46%.
   - Quality choices 4 years ago affect current market share.

Supply estimation:

1. Additional visit per patient costs the hospice around $192.
   - Includes staff wages, cost of medical supplies and operation.
2. Also estimate per-patient non-visit cost and per-period operating cost using reported cost data.

Counterfactuals: reputation persistence

Use structural model to simulate impact of increasing persistence of reputation (τ).

Persistence of reputation $\uparrow \implies$ Equilibrium Quality $\uparrow$

- Mimics policies like online review sites that make quality information widely available and easier to find.
- Review websites like Hospice Compare can improve equilibrium quality at low cost.

Counterfactuals: Medicare prices

Complaint from hospices: "Medicare reimbursement rates too low, not keeping up with inflation."

Use structural model to simulate equilibrium hospice quality as Medicare rates increases.

Medicare reimbursement $\uparrow \implies$ Equilibrium Quality $\uparrow$

- Furthermore, may incentivize new hospice entry and greater competition.
- Higher quality from higher hospice reimbursement can switch patients from ineffective yet expensive end-of-life care.

Counterfactuals: contract design

Growing trend of value-based pricing. What if we partly tied hospice reimbursement to hospice quality?

Following contract structures all achieve 0.78 visits-per-patient-day in equilibrium:

Per-day rate	Per-visit rate	Medicare Spending
223.7	0.0	1.0
142.4	80.0	0.91
61.0	160.0	0.83

Partially tying reimbursement to quality $\rightarrow$ same equilibrium quality at lower Medicare spending.

- However, this reduces hospice profit, may induce exit.

Contributions

1. Contribute to a very sparse literature on hospices.
2. First to estimate structural model of reputation accumulation through quality choice.
   - Method can be used when user-review data is absent.
3. Importance of reputation for i) patients choosing medical providers and ii) medical providers choosing quality.

Conclusion

1. Reputation $\rightarrow$ consumer choice and hospice quality.
2. Build structural model of reputation accumulation through quality choice, recover hospice cost function.
3. Policy counterfactuals:
    - Persistence of reputation $\uparrow$ and Medicare prices $\uparrow \implies$ hospice quality $\uparrow$.
    - A hybrid per-day and per-visit reimbursement scheme achieves the same quality as the current per-day scheme at lower Medicare spending.